

Lecture 9: Disjoint Sets / Union-Find

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601.433/633 Introduction to Algorithms

Slides by Mike Dinitz

Introduction

Informal: Universe of elements, want to maintain *disjoint sets*.

Slightly more formally:

- ▶ **Make-Set(x)**: create a new set containing just x (i.e., $\{x\}$)
- ▶ **Union(x, y)**: Replace set containing x (S) and set containing y (T) with single set $S \cup T$
- ▶ **Find(x)**: Return *representative* of set containing x

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Rules: every set has a *unique* representative.

- ▶ If $\mathbf{x}$ and $\mathbf{y}$ are in same set, $\text{Find}(\mathbf{x}) = \text{Find}(\mathbf{y})$
- ▶ If $\mathbf{x}$ and $\mathbf{y}$ are in different sets, then $\text{Find}(\mathbf{x}) \neq \text{Find}(\mathbf{y})$
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Note: disjoint (and partition) by construction!

Introduction (II)

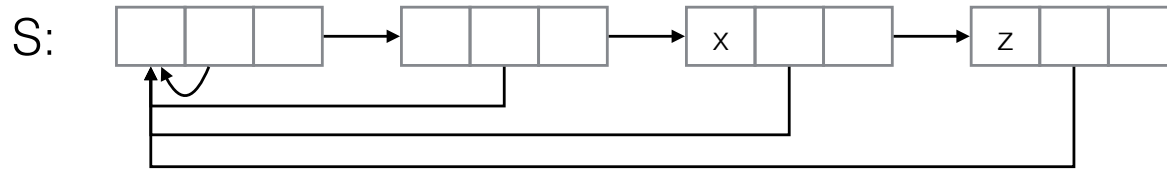
Notation and Notes:

- ▶ **m** operations total
- ▶ **n** of which are Make-Sets (so **n** elements)
- ▶ Assume have pointer/access to elements we care about (like last class)

First Approach: Lists

Linked list for each set.

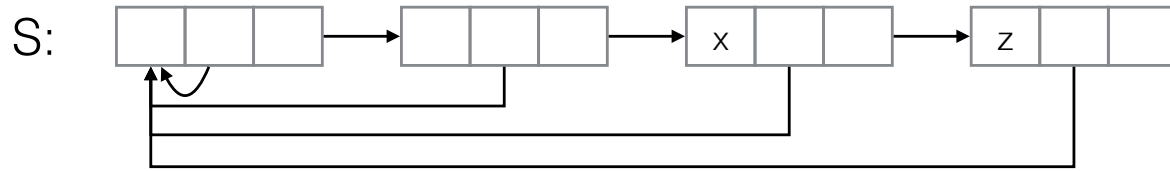
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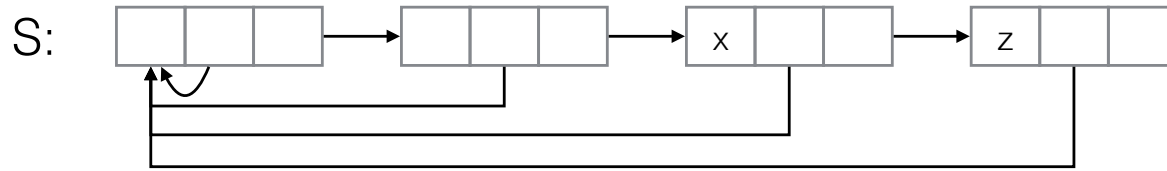
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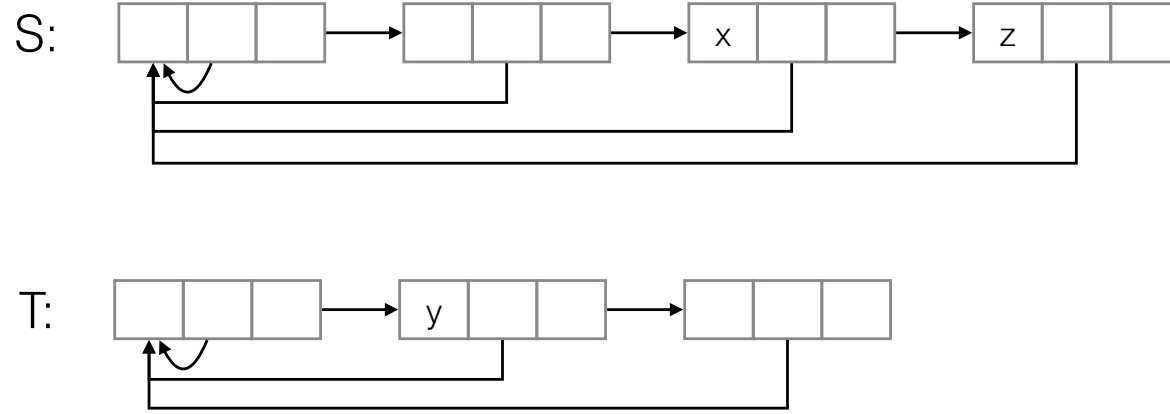


Make-Set(x):

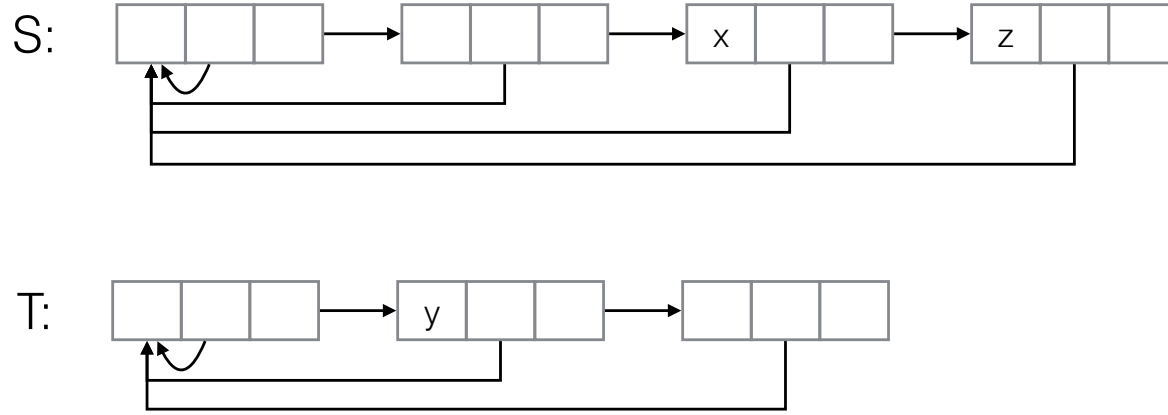


Find(x): return $x \rightarrow \text{head}$

Union(x, y)



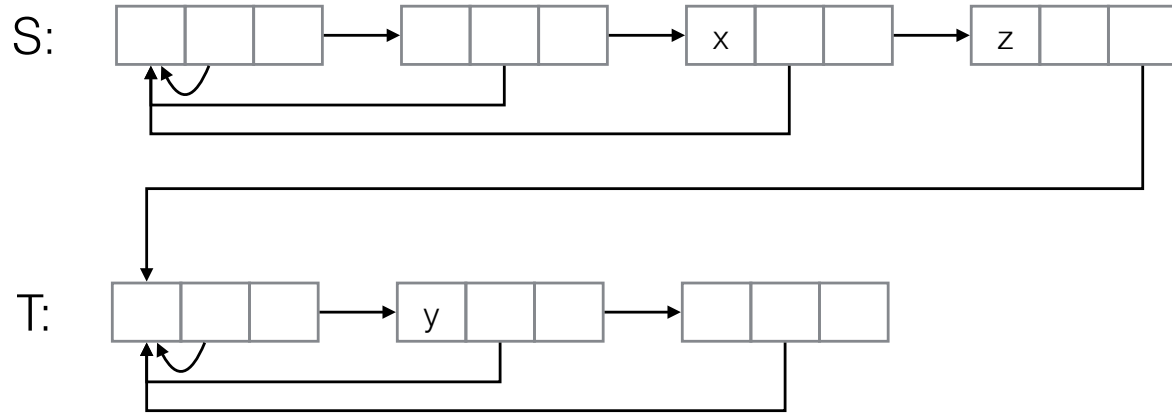
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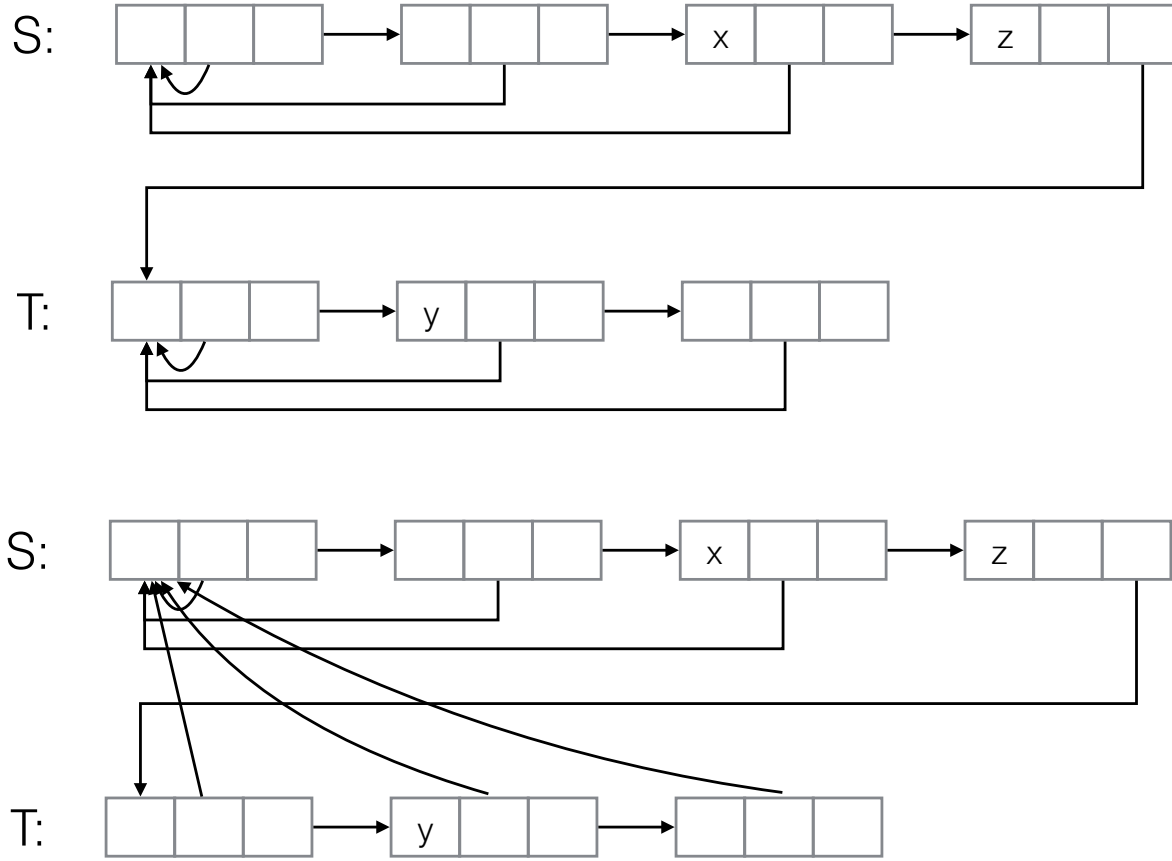
Obvious approach:

- ▶ Walk down **S** to final element **z** (starting from **x**)
- ▶ Set **z** $\rightarrow$ next = **y** $\rightarrow$ head
- ▶ Walk down **T**, set every element's head pointer to **x** $\rightarrow$ head

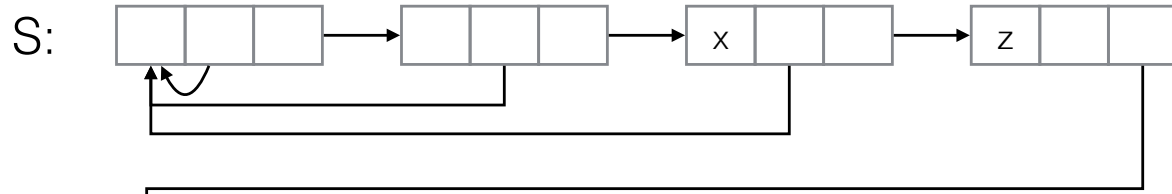
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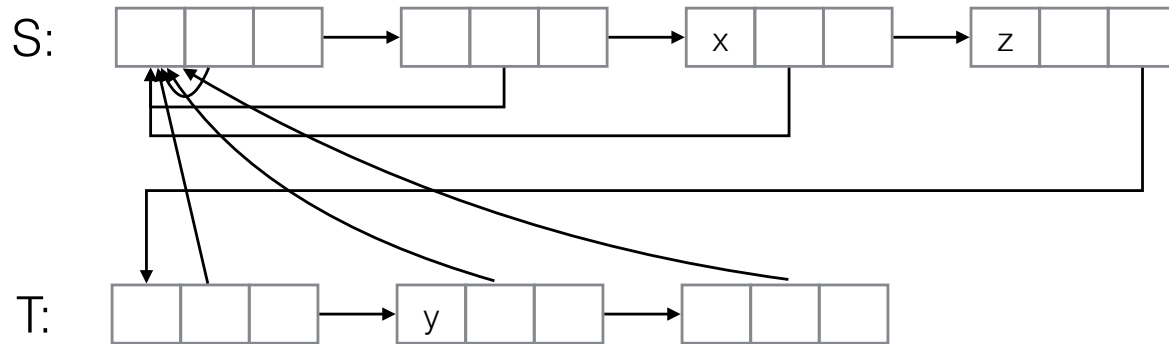
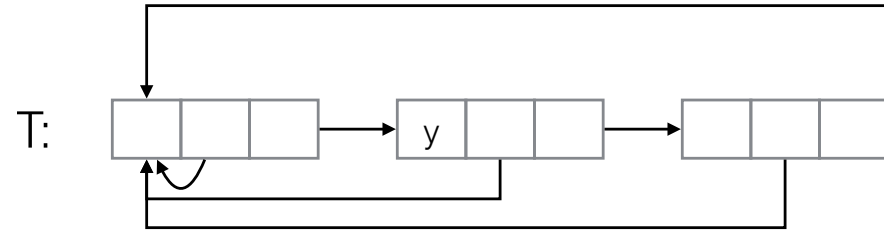
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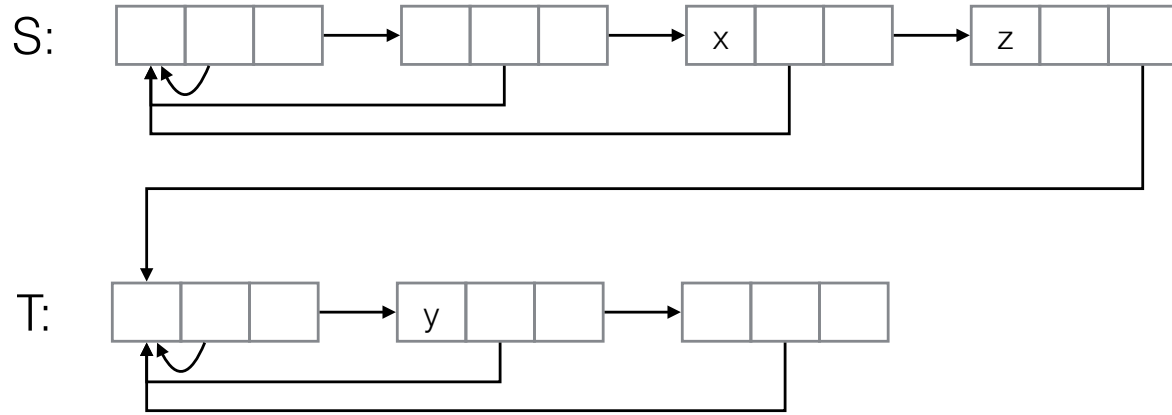
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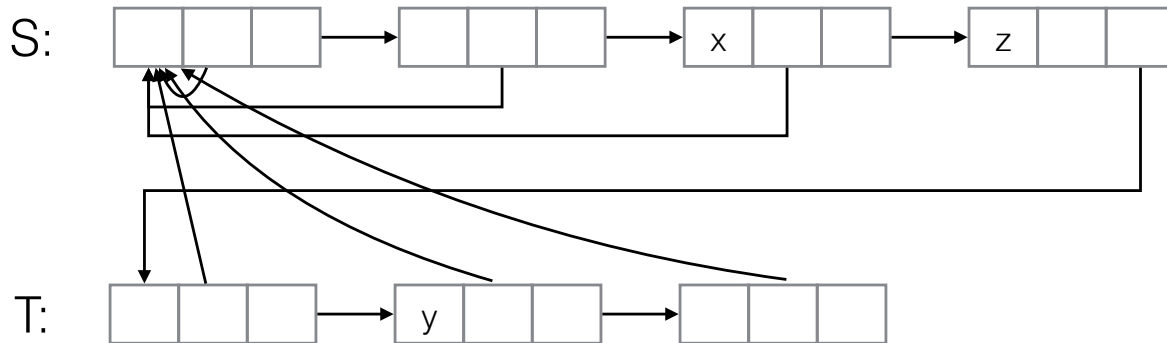
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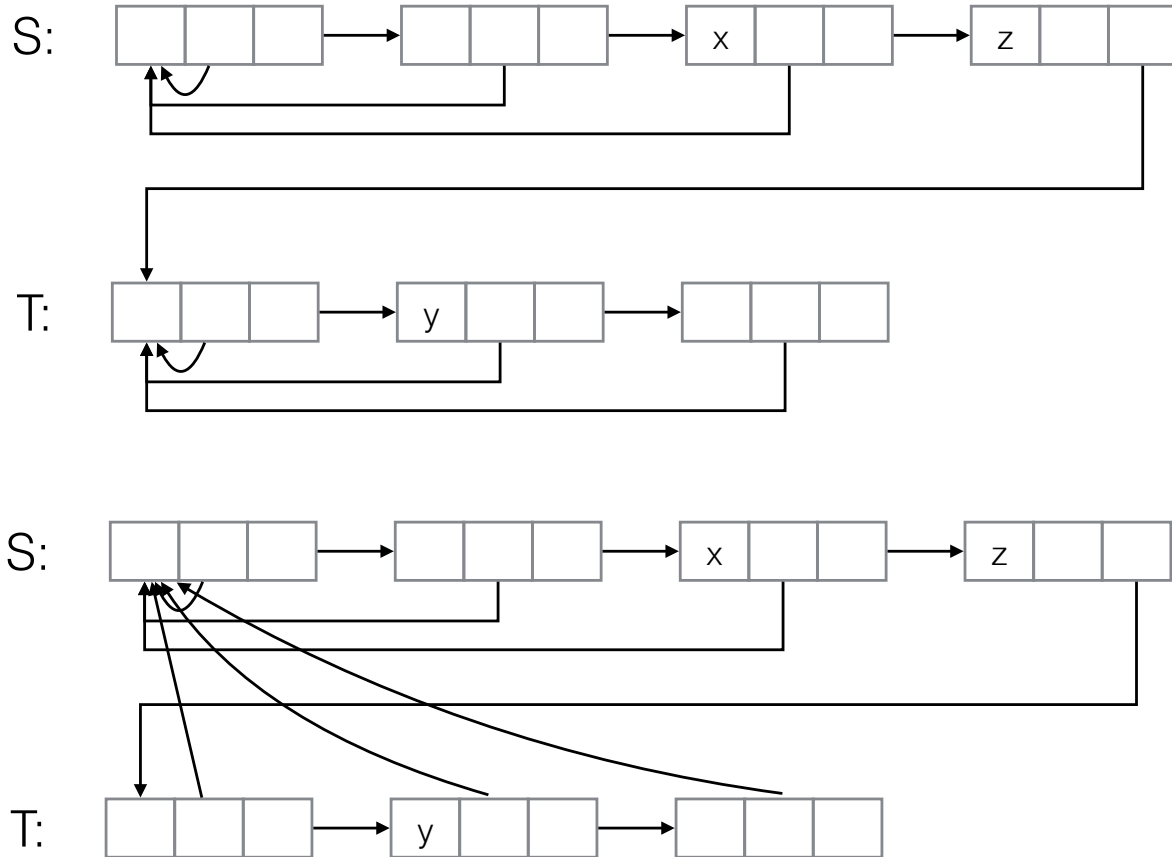
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Running time: $O(|S| + |T|)$



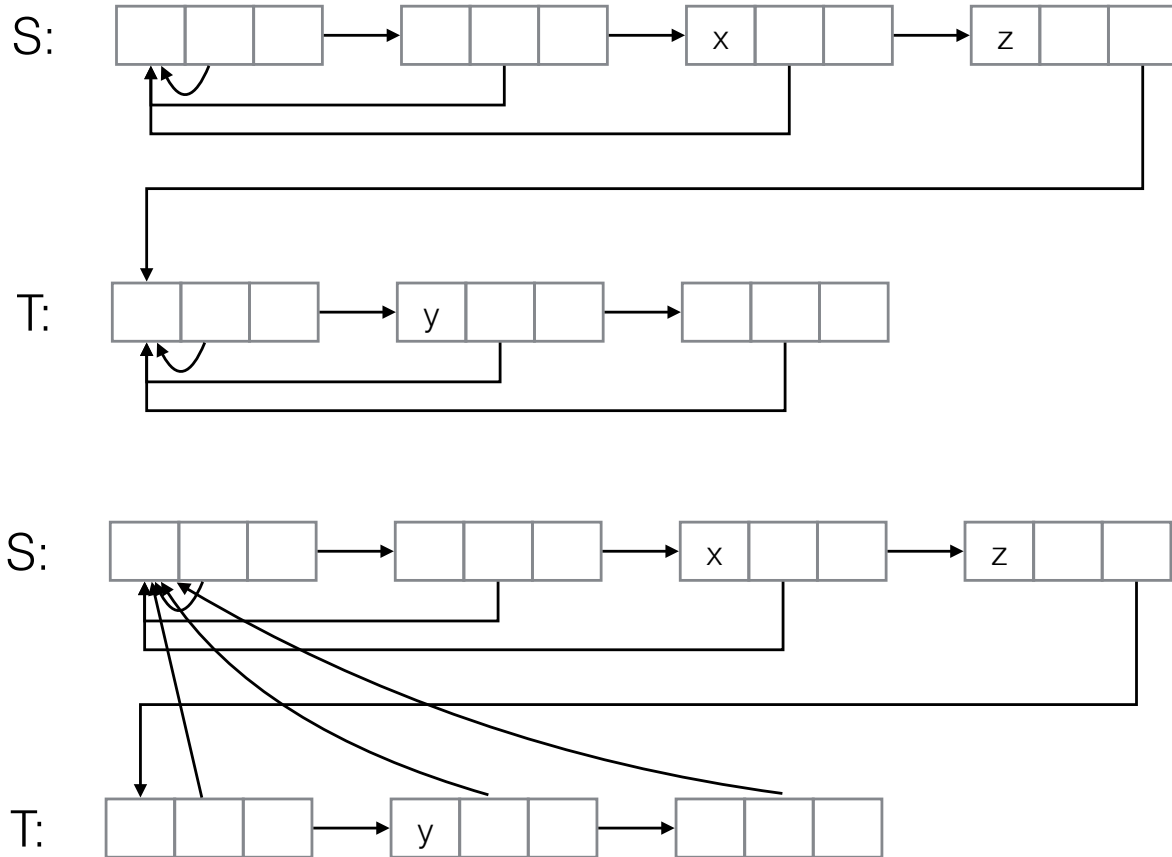
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Since $|S|, |T|$ could be $\Theta(n)$,
can only say $O(n)$ for Unions

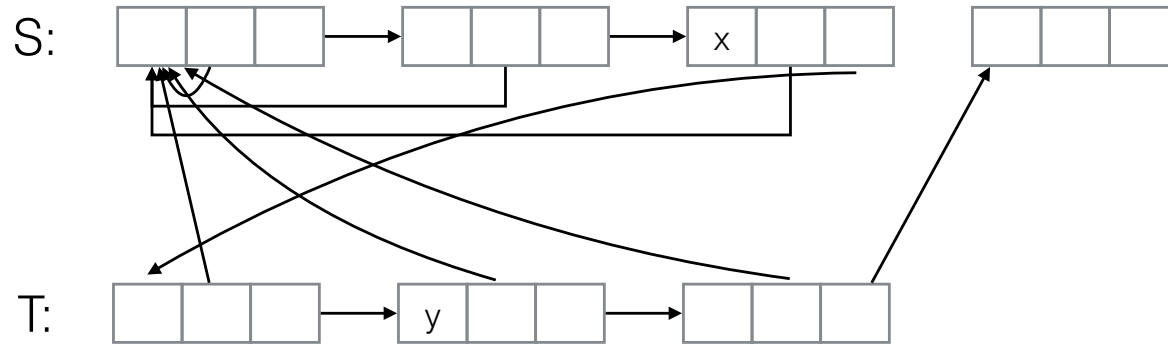
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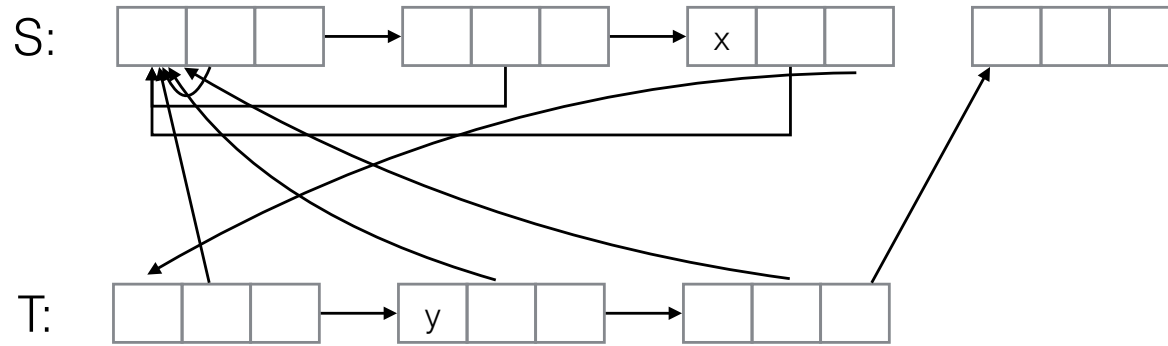
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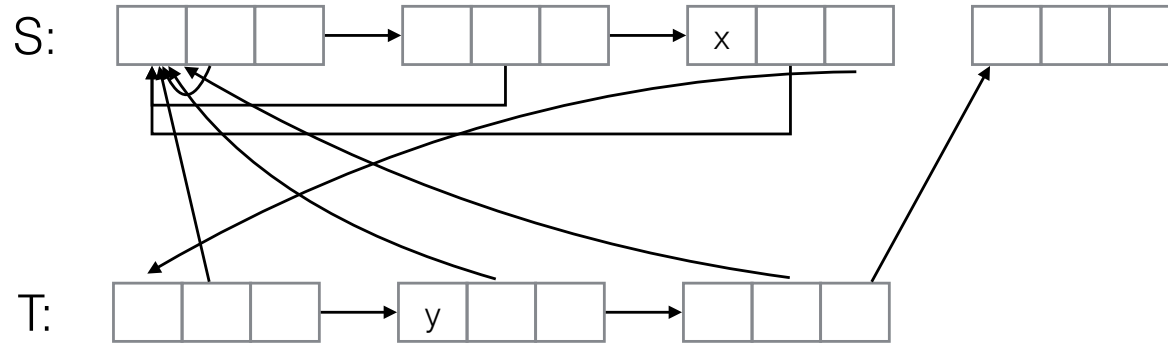


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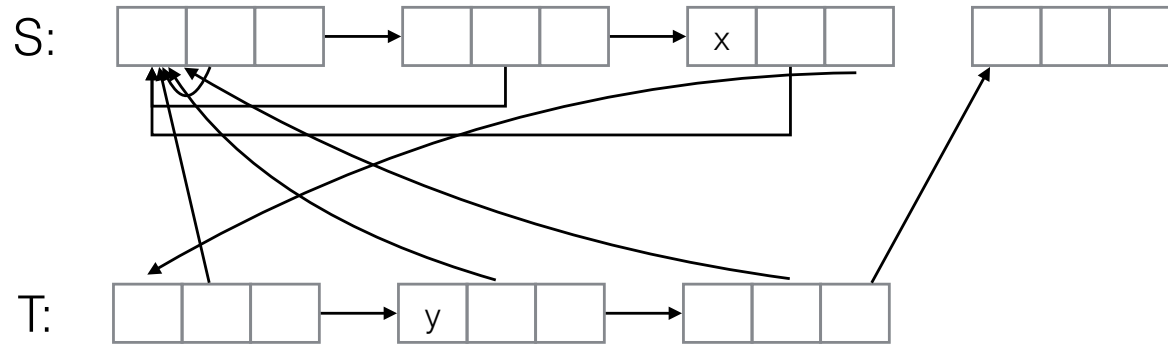


Running time: $O(|T|)$

Improved Union(x, y)

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Running time: $O(|T|)$

- ▶ Still can't say anything better than $O(n)$

Even more improved Union(x, y)

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- ▶ Store size of set in head node.
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Theorem

The amortized cost of Find and Union is $O(1)$, and the amortized cost of Make-Set is $O(\log n)$.

Corollary

The total running time is $O(m + n \log n)$.

Amortized Analysis of List Algorithm

Banking/accounting argument: bank for every element

- ▶ When an element is created (via Make-Set), add **$\log n$** tokens to its bank
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- ▶ When in smaller set of a Union.
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- ▶ Can only happen at most **$\log n$** times.



Amortized Analysis of List Algorithm (cont'd)

Make-Set:

- ▶ True cost: $O(1)$
- ▶ Change in banks: $\log n$

$\Rightarrow$ Amortized cost: $O(1) + O(\log n) = O(\log n)$

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Find:

- ▶ True cost: $O(1)$
- ▶ Change in banks: 0

⇒ Amortized cost: $O(1) + 0 = O(1)$

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Find:

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Union:

- ▶ True cost: $\min(|S|, |T|)$
- ▶ Change in banks: $-\min(|S|, |T|)$

⇒ Amortized cost: $\min(|S|, |T|) - \min(|S|, |T|) = 0 = O(1)$.

Even Better

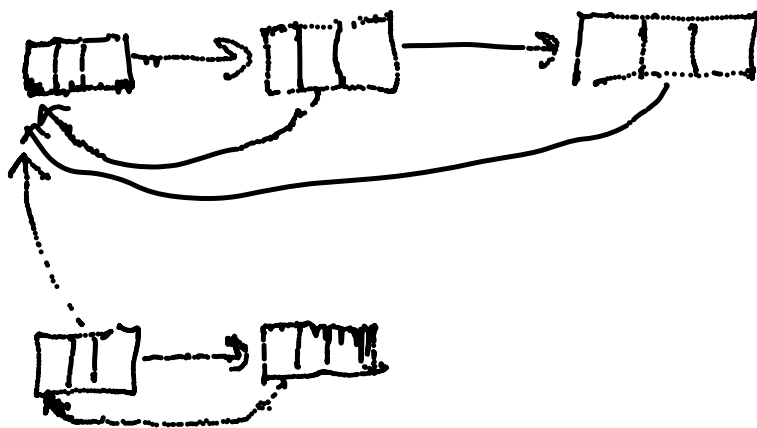
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Idea 2: *Union By Rank*

- ▶ Size of set was important for lists, less important for trees.
- ▶ Choose which set to splice into which by *rank* of trees (related to height)

Main Result

Theorem

When using Path Compression and Union By Rank, total time at most $O(m \log^ n)$.*

$\log^*$: iterated $\log_2$.

- ▶ $\log^* n = \#$ times apply $\log_2$ until get to ≤ 1

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Stronger theorem: total time at most $O(m \cdot \alpha(m, n))$.

- ▶ $\alpha(m, n)$: inverse Ackermann function. Grows even slower than $\log^*$.
- ▶ See CLRS for details

Formal Procedures: Make-Set and Find

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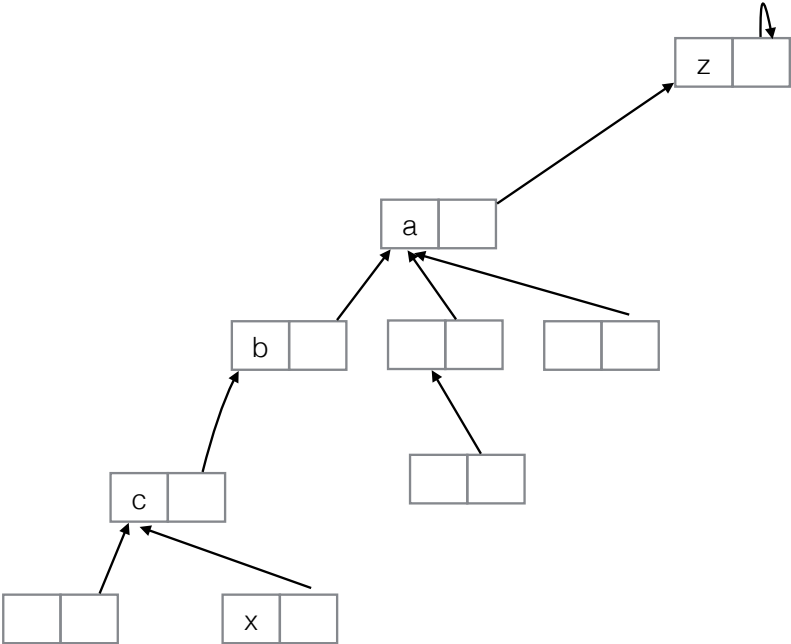
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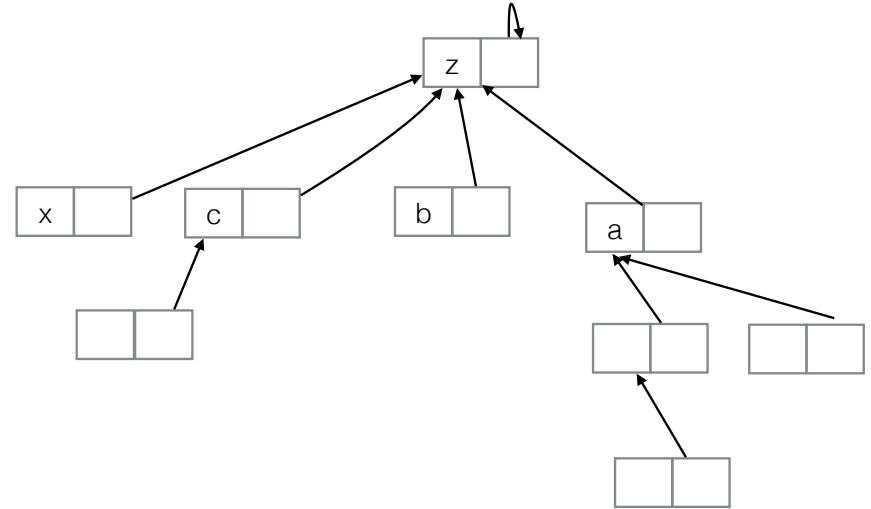
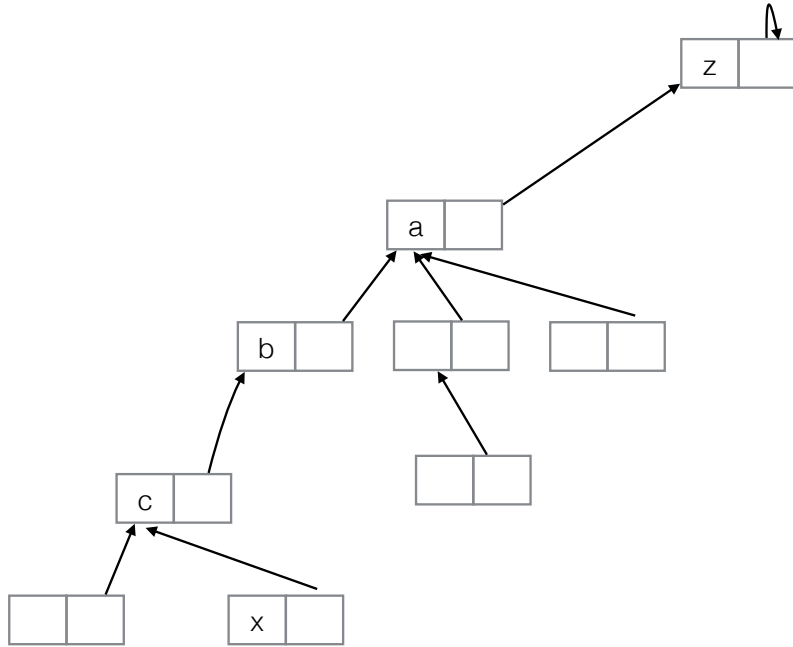
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Running time of Find: depth of x (distance to root)

Find example



Find example



Formal Procedure: Union

Link(r_1, r_2): Only applied to root nodes

- ▶ If $r_1 \rightarrow \text{rank} > r_2 \rightarrow \text{rank}$, set $r_2 \rightarrow \text{parent} = r_1$
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Union(x, y): Link(Find(x), Find(y))

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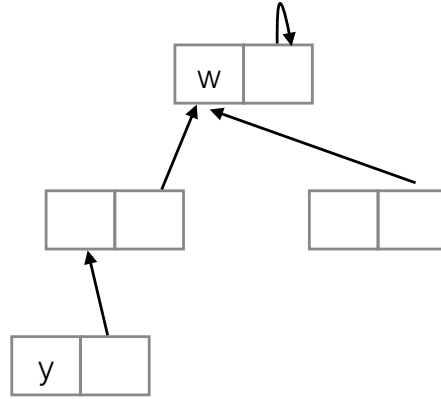
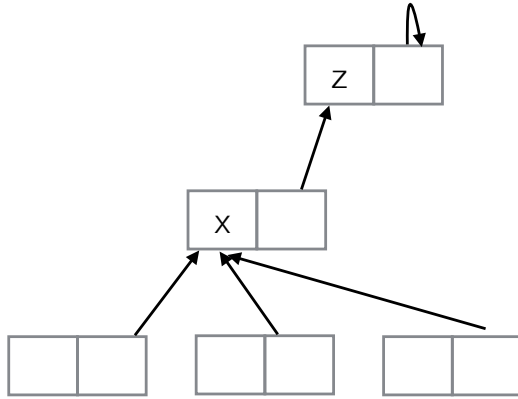
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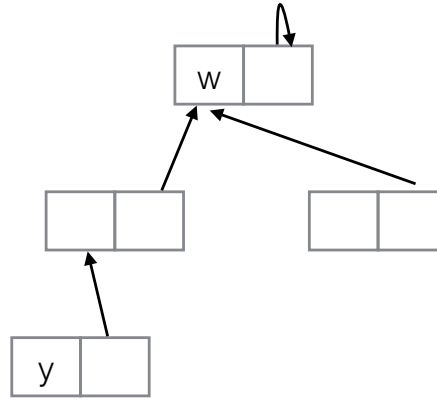
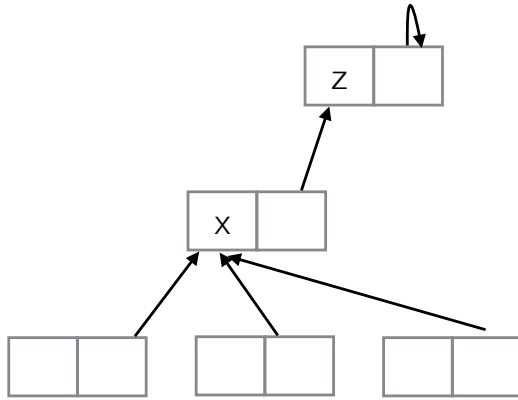
Union(x, y): Link(Find(x), Find(y))

- ▶ Running time: $\text{depth}(x) + \text{depth}(y)$

Union example

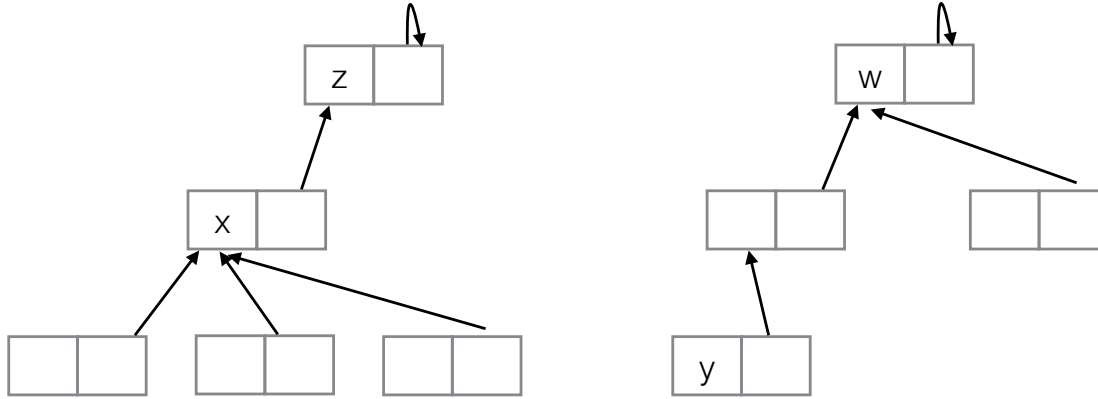


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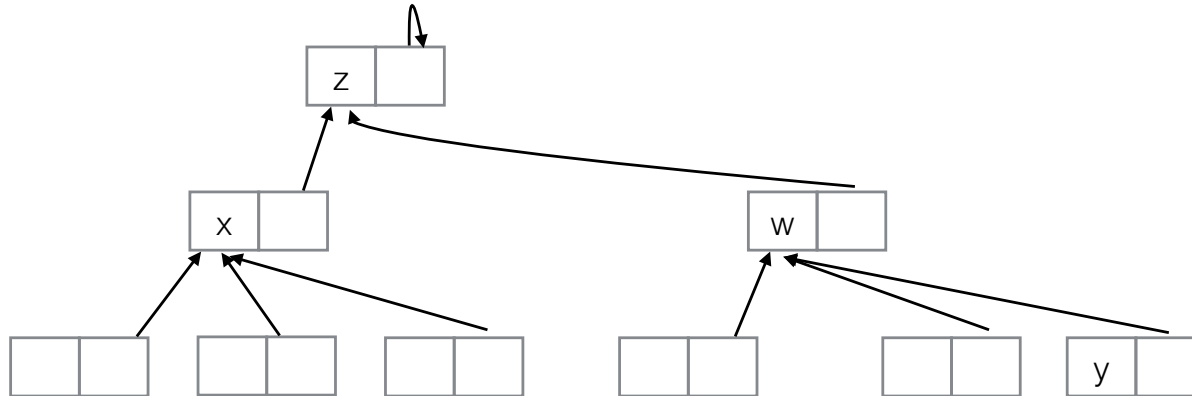


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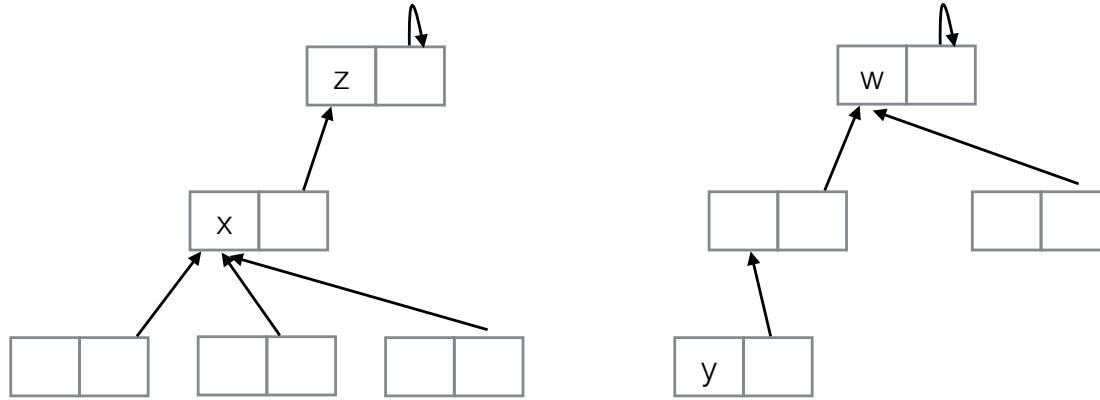
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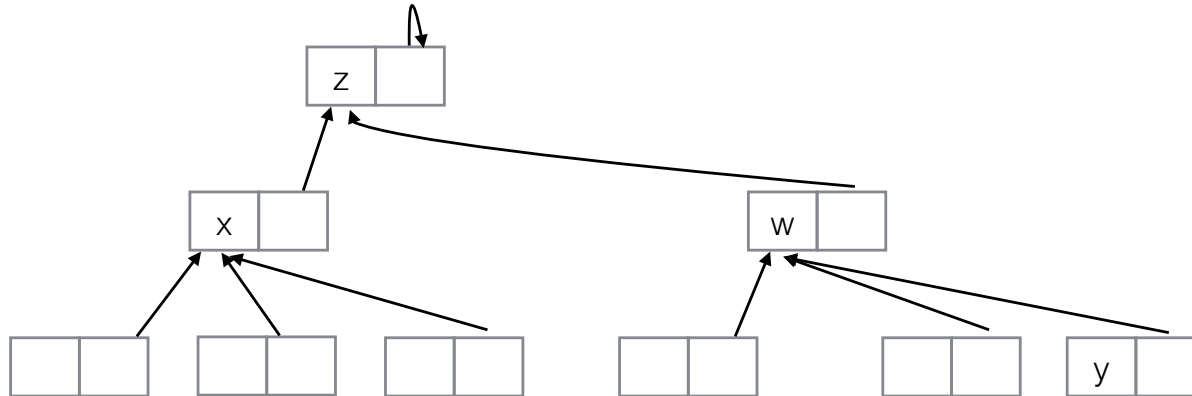


Union example



If $z \rightarrow \text{rank} \geq w \rightarrow \text{rank}$

If $z \rightarrow \text{rank} = w \rightarrow \text{rank}$,
then $(z \rightarrow \text{rank})++$



Properties of Ranks

1. If x not a root, then $(x \rightarrow \textit{rank}) < (x \rightarrow \textit{parent} \rightarrow \textit{rank})$
2. When doing path compression, if parent of x changes, new parent has rank strictly larger than old parent
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4. When x first reaches rank r , there are at least 2^r nodes in tree rooted at x .

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⇒ At least $2^{r-1} + 2^{r-1} = 2^r$ nodes in combined tree.



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Lemma

There are at most $n/2^r$ nodes of rank at least r .

Proof.

Let x be node of rank at least r . Let S_x be descendants of x when it first got rank r .
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So at most $2m$ Finds, want to bound total # parent pointers followed.

- ▶ At most one parent pointer to root per Find $\implies$ at most $O(m)$ parent pointers to roots.
- ▶ So only need to worry about parent pointers to non-roots.

Main Result II: Buckets

Put elements in buckets according to rank (only in analysis).

Notation: $2 \uparrow i$ denotes a tower of i 2's

- ▶ $2 \uparrow 1 = 2$, $2 \uparrow 2 = 2^2 = 4$, $2 \uparrow 3 = 2^{2^2} = 2^4 = 16$, $2 \uparrow 4 = 2^{2^{2^2}} = 2^{16} = 65536$
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From Lemma: at most $n / (2^{2 \uparrow (i-1)}) = n / (2 \uparrow i)$ elements in bucket i .

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$$\begin{aligned} \sum_x \alpha(x) &= \sum_{i=1}^{O(\log^* n)} \sum_{x \in B(i)} \alpha(x) \leq \sum_{i=1}^{O(\log^* n)} \sum_{x \in B(i)} (2 \uparrow i) \leq \sum_{i=1}^{O(\log^* n)} \frac{n}{2 \uparrow i} (2 \uparrow i) = O(n \log^* n) \\ &\leq O(m \log^* n) \end{aligned}$$